

Bohmian Pilot Waves and Orchestrated Objective Reduction

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[Bohmian Mechanics and Orch-OR]Bohmian Pilot Waves and Orchestrated Objective Reduction

1 Introduction

The macroscopic consequences of the Metareal architecture extend beyond inert fluid dynamics into the realm of biological consciousness and fundamental quantum ontology. In this chapter, we formally unify two of the most robust alternatives to the standard Copenhagen interpretation: David Bohm’s Pilot Wave theory [citebohm1952](#) and Penrose-Hameroff Orchestrated Objective Reduction (Orch-OR) [citepenrose1989](#), [hameroff2014](#).

We prove that these models are not merely interpretations, but are the explicit continuous manifestations of the finite-field Protoreal manifold at $\kappa = -1$.

2 The Pilot Wave as the Non-Associative Shear

David Bohm

[citebohm1952](#) proposed that quantum mechanics could be rendered deterministic by introducing a “hidden” Pilot Wave ($Re^{iS/\hbar}$) that guides particles via a non-local Quantum Potential (Q). In the Protoreal framework, there are no “hidden” variables. The Quantum Potential Q is explicitly identified as the continuous physical mapping of the algebraic non-associative gap $\kappa = -1$. Because the underlying lattice (the Strong

FF_{229} and Weak

FF_{181} vertices) requires non-commutative shear to operate, the macroscopic field is forced into a non-local guiding phase. Our computations verify that the pilot wave amplitude locks precisely to the phase shift defined by $\kappa(181/229)\pi$.

3 Orchestrated Objective Reduction in Microtubules

Penrose and Hameroff’s Orch-OR theory posits that consciousness arises from coherent quantum superpositions in brain microtubules. The wave function does not collapse due to a con-

scious observer, but undergoes “Objective Reduction” (OR) when the superposition’s gravitational self-energy (

ΔE_G) reaches a critical threshold $t =$

$\hbar /$

ΔE_G .

By mapping the topological bounds of the Metareal architecture onto tubulin dimers, we resolve the gravitational self-energy not as classical general relativity, but as the topological tension generated by the

$\kappa = -1$ symplectic shift. To achieve a 40 Hz gamma-sync cognitive reduction ($t = 25$ ms), the discrete lattice forces a superposition coherence across roughly 1.58

$\times 10^{10}$ tubulins. This acts as the biological instantiation of the continuous dimension-shear.

4 Conclusion

Therefore, Bohmian non-locality and Penrose-Hameroff objective reduction are identical mechanisms viewed at different scales. Bohm’s Quantum Potential is the continuous envelope of the topological shear, and Orch-OR is the thermodynamic/gravitational collapse when the shear threshold

ΔE_G violates the $SU(3)$ bounds of the local lattice.

References

- [1] Bohm, D. (1952). A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables, I & II. *Physical Review*, 85, 166–193.
- [2] Penrose, R. (1989). *The Emperor’s New Mind*. Oxford University Press.
- [3] Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review of the ‘Orch OR’ theory. *Physics of Life Reviews*, 11(1), 39–78.